

GANESHKUMAR V

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Education

Indian Institute Technology Gandhinagar

PH.D. MECHANICAL ENGINEERING

CPI : 9.61/10 | Prime Minister's Research Fellow

Gujarat

Aug. 2020 - present

Anna University - Madras Institute of Technology (MIT)

B.E MECHANICAL ENGINEERING

CGPA : 9.84/10 | University Gold Medalist for highest academic performance

Chennai

Aug. 2016 - May 2020

Velankanni Matriculation Higher Secondary School

HIGHER SECONDARY EDUCATION- TAMIL NADU STATE BOARD

Percentage : 96.67 % | School topper

Chennai

Jun. 2014 - May 2016

Velankanni Matriculation Higher Secondary School

SECONDARY EDUCATION - TAMIL NADU STATE BOARD

Percentage : 98.2 % | 100 % in Maths, Science and Social Science

Chennai

Jun. 2013 - May 2014

Research/Academic Projects

Ph.D. Research: Computational Study on Propulsive Performance of Metal-Based Refrigerated and Gelled Composite Propellants

IIT Gandhinagar

SUPERVISOR: PROF. DILIP S SUNDARAM

Dec. 2020 - Present

- Propulsion Research:
 - Developed a **multiphase CFD framework** and implemented it in **OpenFOAM** to accurately capture the **multiphase flow dynamics in solid rocket motors** with metal-based propellants.
 - The developed novel OpenFOAM based CFD solver was optimized and used to **quantify the propulsive performance such as thrust and specific impulse** for solid rocket motors with metal-based propellants.
- Multiphase Combustion Research:
 - Developed and implemented a **multiphase combustion CFD framework in OpenFOAM ("regressionFoam")** to replicate strand burner experiments and **predicted the burning rates** of metal particles in liquid oxidizer mixtures.
 - Identified key factors affecting combustion performance and **formulated a hypothesis** to resolve discrepancies between experimental results and theoretical predictions.

Computational Performance Evaluation and Optimization of Combustion Chemistry Calculations in OpenFOAM

IIT Gandhinagar

SUPERVISOR: PROF. DILIP S SUNDARAM

Summer 2024

- Profiled **OpenFOAM's combustion libraries** and solver using **Intel VTune Profiler**, identifying major computational bottlenecks in gas-phase combustion simulations.
- Implemented **Hybrid MPI-OpenMP parallelization** strategies to accelerate chemical source term evaluations, achieving a **208x speedup** in computational performance.

Numerical Modelling of Electrohydraulic Forming Process

MIT - Anna University

SUPERVISOR: PROF. P. GANESH

Nov. 2019 - May 2020

- Conducted coupled **Smoothed Particle Hydrodynamics (SPH)** and Finite Element Method (FEM) simulations in ABAQUS to accurately model the **electro-hydraulic forming** of sheet metal.

Determination of Acoustic Resonance Modes in the Air Cavity

IIT Madras

SUPERVISOR: PROF. K. SRINIVASAN

May 2019 - Jul. 2019

- Conducted **Modal Acoustic Analysis** in ANSYS to determine the **resonance frequencies** of an air cavity resembling a rocket combustion chamber and validated results against experimental data.

Publications

- 2025 **Journal of Propulsion and Power** , *Computational Study of Propulsive Performance of Frozen Nano-Aluminum and Water (ALICE) Mixtures*, Ganeshkumar V and Dilip S Sundaram, [Link](#)
- 2025 **AIAA SciTech Forum 2025** , *Effects of Particle Size and Mixture Composition on Propulsive Performance of Gelled and Refrigerated Aluminum-Liquid Oxidizer Propellants*, Ganeshkumar V and Dilip S Sundaram, [Link](#)
- 2024 **Combustion And Flame** , *A Computational Fluid Dynamics Model of Combustion of Nanoaluminum-Water Propellant Strands*, Prasanna Kulkarni, Ganeshkumar V and Dilip S Sundaram (Accepted - Mar 2025)

Academic Achievements

- 2022 **Prime Minister's Research Fellowship (PMRF) - Government of India** , Awarded for conducting PhD research at IIT Gandhinagar
- 2020 **University Gold Medal - Anna University** , Secured highest CGPA among all the students in the Department of Mechanical Engineering
- 2019 **IIT Madras Summer Fellowship Programme** , Ranked among the top 20 students nationwide
- 20'17-20 **RNTBCI - AU Scholarship** , Awarded prestigious scholarship for meritorious students by Renault Nissan Technology & Business Centre India (RNTBCI)

Skills

Programming Skills	Object Oriented Programming, MPI, OpenMP, CUDA
Programming Languages	C++, Python, MATLAB
Software Tools	OpenFOAM, Ansys-Fluent, ABAQUS

Relevant Courses

- 2023 **Combustion** , IIT Gandhinagar, Grade: 10/10
- 2022 **Aircraft and Rocket Propulsion** , IIT Gandhinagar, Grade: 10/10
- 2021 **Advanced Computational Fluid Dynamics** , IIT Gandhinagar, Grade: 10/10
- 2020 **Computational Fluid Dynamics** , IIT Gandhinagar, Grade: 10/10

Experience

Graduate Teaching Fellow

INDIAN INSTITUTE OF TECHNOLOGY GANDHINAGAR AND NPTEL

Gujarat & Online

Aug. 2022 - Present

- Independently tutored students in both fundamental and advanced topics, including **Computational Fluid Dynamics**.
- Collaborated with instructors and successfully led over six courses as a **lead Teaching Assistant**.
- Collaborated with various **IIT professors** and conducted weekly tutorial sessions for multiple **NPTEL** courses. [YouTube](#)

Summer Research Intern

INDIAN INSTITUTE OF TECHNOLOGY MADRAS UNDER PROF. K. SRINIVASAN

Chennai

May. 2020 - Jul. 2020

- Performed numerical analysis to determine the resonance modes in the air cavity and validated with the experimental results